Christian González Pérez

Aerospace engineer | Simulation, control and scientific software

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Profile

Aerospace engineer interested in building software tools for engineering problems, combining simulation, control, scientific software and machine learning. Academic and project experience in aerial vehicles, meteorology and space systems. I turn technical needs into applications, from defining requirements to integrating and checking results, with the support of coding agents.

Education

Bachelor's degree in Aerospace Engineering 2022–2026

University of León, Spain | Average grade: 9.17/10. Extraordinary End-of-Degree Award, 2025–2026 academic year.
Undergraduate thesis: cascaded hybrid quadcopter control with neural replacement of the outer loop. Grade: 9.5/10.

University microcredential in Model-Based Engineering 2026

University of León | 5 ECTS credits. Systems modelling with Capella.

Flight Simulators Course (Aeroplanes) 2025–2026

University of León | 130 hours, 6 ECTS credits. Visual and instrument flight, navigation and flight planning.

Technical projects

Neural control of a quadcopter – undergraduate thesis 2026

Repository | Project and thesis

- Designed a modular six-degree-of-freedom simulator in Python to compare classical controllers and MLP, GRU and LSTM networks for trajectory tracking.
- Defined the physics and control architecture, designed experiments and analysed tracking data, errors and telemetry to compare controller behaviour on known and unseen trajectories.

SkyBrief – aeronautical weather 2026

Repository

- Developed an academic prototype for retrieving METAR and TAF reports, weather maps and route profiles for over 700 European airports.
- Integrated weather reports and historical ERA5 episodes with risk indicators, maps and route profiles.

Technical projects (continued)

FireSat25 / ULEsat25 – CubeSat mission 2025–2026

- Contributed to the academic design of a wildfire observation mission as part of a four-person onboard data handling (OBDH) team.
- The team proposed an onboard computer architecture and built a Python client-server demonstrator for simulated telemetry and telecommands.

Vibration analysis of a drone propeller 2026

Modal and harmonic response analysis in ANSYS, studying frequencies and damping and post-processing results in Python.

Leadership and other activities

Founder and president, Club de Moda ULe 2024–2026

Founded and led a university association of around 15 active members, coordinating its team, activities and relationships with the university and external partners. Together with the team, organised Campus Runway, a fashion show bringing together over 10 models, four businesses and around 50 attendees. Developed interactive web applications for its activities.

Competitions and team projects 2024–2026

- D.R.A.G.: a conceptual passive aerodynamic braking project for space vehicles, one of the winners of the 2026 ULE Prototype Competition; team project with preliminary simulations.
- Second place with team R.A.R. at EUDIS Defence Hackathon Spring 2026 Spain. Participated in NASA Space Apps Challenge León 2024.

Skills and languages

Engineering and data: flight dynamics and control; data analysis and visualisation; aeronautical meteorology; model-based engineering; modal analysis, vibrations and CFD.
Tools: Python, PyTorch, MATLAB (symbolic computation, control and data analysis), ANSYS, GMAT, SaVoir, Capella, Git and GitHub.
Web development: applications with Astro and React, using JavaScript, HTML and CSS.
AI-assisted development: requirements and specification-driven development (SDD), planning and multi-agent orchestration, cross-review of solutions, and verification of code and behaviour.
Languages: native Spanish; English: Cambridge B2 (2022); comfortable in technical conversations.
Mobility: willing to relocate within Spain or internationally.